

Proofs using Set Identities

Suppose, E is a universal set. Now consider the following seven results for any subsets A, B, C of E .

$$R1: A \cup \emptyset = A$$

$$R2: A \cap A^c = \emptyset$$

$$R3: (A^c)^c = A$$

$$R4: A \cap B = B \cap A$$

$$R5: (A \cap B)^c = A^c \cup B^c$$

$$R6: A \setminus B = A \cap B^c$$

$$R7: (A \cup B) \cap C = (A \cap C) \cup (B \cap C)$$

Now, using only seven results and the properties of "=", prove algebraically the following for arbitrary subsets A, B, C of E .

$$(\alpha) \quad (B \setminus A) \cup (C \setminus A) = (B \cup C) \setminus A$$

$$(\beta) \quad B^c \setminus (B^c \setminus A) = B^c \cap A.$$

Proof of (α) :

$$\begin{aligned} (B \setminus A) \cup (C \setminus A) &= (B \cap A^c) \cup (C \cap A^c) \quad (\text{By } R6) \\ &= (B \cup C) \cap A^c \quad (\text{By } R7) \\ &= (B \cup C) \setminus A \quad (\text{By } R6) \end{aligned}$$

$$\therefore (B \setminus A) \cup (C \setminus A) = (B \cup C) \setminus A \quad \square$$

Proofs using Set Identities

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$$(\beta) \quad B^c \setminus (B^c \setminus A) = B^c \cap A.$$

$$\begin{aligned} \text{Proof of } (\beta): \quad B^c \setminus (B^c \setminus A) &= B^c \setminus (B^c \cap A^c) \quad (\text{By } R6) \\ &= B^c \cap (B^c \cap A^c)^c \quad (\text{By } R6) \\ &= B^c \cap ((B^c)^c \cup (A^c)^c) \quad (\text{By } R5) \\ &= B^c \cap (B \cup A) \quad (\text{By } R3) \\ &= (B \cup A) \cap B^c \quad (\text{By } R4) \\ &= (B \cap B^c) \cup (A \cap B^c) \quad (\text{By } R7) \\ &= \emptyset \cup (B^c \cap A) \quad (\text{By } R2 \text{ and } R4) \\ &= (B^c \cap A) \cup \emptyset \quad (\text{By } R4) \\ &= B^c \cap A \quad (\text{By } R1) \end{aligned}$$

Ex. Let A, B be subsets of the universal set E . Using only the following four results and the properties of "=", prove that $A \cap E = A$.

$$(R1) \quad E^c = \emptyset.$$

$$(R2) \quad A \cup \emptyset = A.$$

$$(R3) \quad (A^c)^c = A.$$

$$(R4) \quad (A \cap B)^c = A^c \cup B^c.$$

$$\begin{aligned} \text{Proof. } A \cap E &= (A \cap E)^c \quad (\text{By } R3) \\ &= (A^c \cup E^c)^c \quad (\text{By } R4) \\ &= (A^c \cup \emptyset)^c \quad (\text{By } R1) \\ &= (A^c)^c \quad (\text{By } R2) \\ &= A \quad (\text{By } R3) \end{aligned}$$

$$\therefore A \cap E = A \quad \square$$

Note:

Basic properties of unions

- $A \cup B = B \cup A$.
- $A \cup (B \cup C) = (A \cup B) \cup C$.
- $A \subseteq (A \cup B)$.
- $A \cup A = A$.
- $A \cup U = U$.
- $A \cup \emptyset = A$.
- $A \subseteq B$ if and only if $A \cup B = B$.

Basic properties of intersections:

Let A and B be two sets. Then

1. $A \cap B = B \cap A$ (Commutativity).
2. $A \cap (B \cap C) = (A \cap B) \cap C$ (Associativity).
3. $A \cap B \subseteq A$ and $A \cap B \subseteq B$
4. $A \cap A = A$, $A \cap U = A$.
5. $A \cap \emptyset = \emptyset$.
6. $A \subseteq B$ if and only if $A \cap B = A$.

Some basic properties of complements:

- $A \setminus B \neq B \setminus A$ for $A \neq B$.
- $A \cup A' = U$.
- $A \cap A' = \emptyset$.
- $(A')' = A$.
- $\emptyset \setminus A = \emptyset$.
- $A \setminus \emptyset = A$.
- $A \setminus A = \emptyset$.
- $A \setminus U = \emptyset$.
- $A \setminus A' = A$ and $A' \setminus A = A'$.
- $U' = \emptyset$ and $\emptyset' = U$.
- $A \setminus B = A \cap B'$.
- if $A \subseteq B$ then $A \setminus B = \emptyset$.

Proofs left as an ex.

(use (i)) $A \subseteq B \stackrel{\text{def}}{\Leftrightarrow} \forall x, x \in A \Rightarrow x \in B$
(iii) $A = B \stackrel{\text{def}}{\Leftrightarrow} \forall x, x \in A \Leftrightarrow x \in B$

Counter ex

Venn diagram proof

Two important propositions:

They state that, for sets A, B, C

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$$

$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$$

When A is a universe to which B and C belong, De Morgan's laws can be stated more simply as,

$$(B \cup C)' = B' \cap C'$$

$$(B \cap C)' = B' \cup C'$$

In words:

"The complement of a union is the intersection of the complements." "The complement of an intersection is the union of the complements."

Part 1 $A - (B \cup C) = (A - B) \cap (A - C)$:

$$\begin{aligned} A - (B \cup C) &= A \cap (B \cup C)^c \\ &= A \cap (B^c \cap C^c) \quad \swarrow \downarrow ? \\ &= (A \cap B^c) \cap (A \cap C^c) \\ &= (A - B) \cap (A - C) \end{aligned}$$

$$(A = A \cap A)$$

Mid-Exam 02

Q1 Mathematical Induction version I

Q2 " " " Version II

Q3 Set Theory

Q4 " "

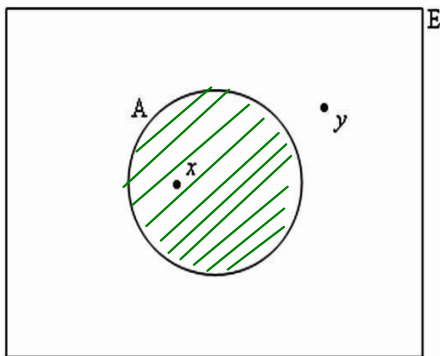
* proof by contraposition.
* " " Contradiction.
* Direct proof.

3.4 Venn diagrams for sets

3.4(a) Venn diagrams for two sets.

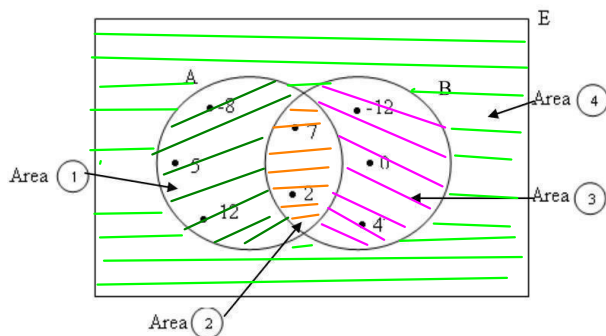
A Venn diagram is a pictorial representation of sets where sets are represented by enclosed areas in the plane. The universal set E is represented by the points in a rectangle, and the other sets are represented by closed circular (disks) or elliptical figures lying within the rectangle.

Thus for example in the following diagram $x \in A$ but $y \notin A$.



Example: Let $A = \{-8, 2, 5, 7, 12\}$ and $B = \{-12, 0, 2, 4, 7\}$.

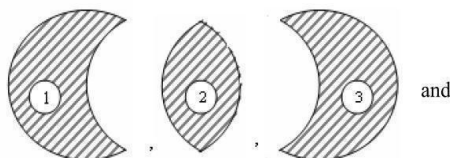
We can represent A, B in a Venn diagram in the following way:



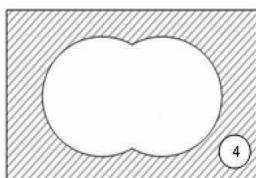
We notice that there are four areas. We have labeled them (1), (2), (3),

(4)

i.e.



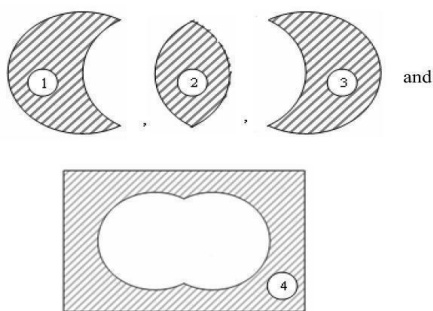
and



We notice that there are four areas. We have labeled them (1), (2), (3), (4).

(4)

i.e.



(1) An object x in area (1), if and only if, $x \in A$ and $x \notin B$

(2) An object x in area (2), if and only if, $x \in A$ and $x \in B$

(3) An object x in area (3), if and only if, $x \notin A$ and $x \in B$

(4) An object x in area (4), if and only if, $x \notin A$ and $x \notin B$

Note: (1) Area (2) corresponds to $A \cap B$

(2) Area (1), (2), (3) corresponds to $A \cup B$

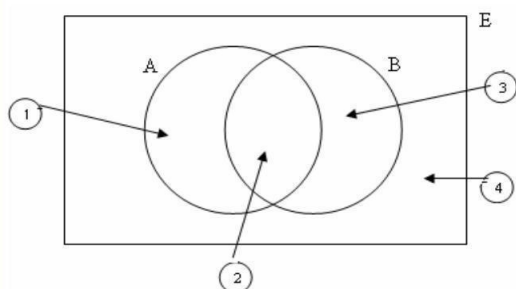
(3) Area (1) corresponds to $A \setminus B$

(4) Area (3) corresponds to $B \setminus A$

(5) Area (4) corresponds to the collection of all objects x such that $x \notin A$ and $x \notin B$. Usually we have a universal set E and area (4) corresponds to all objects in E such that $x \notin A$ and $x \notin B$.

General Venn diagrams for two sets.

Let us now consider two arbitrary sets A, B . We draw the Venn diagram as:



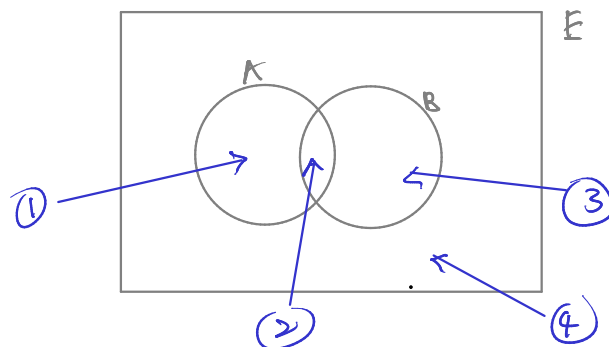
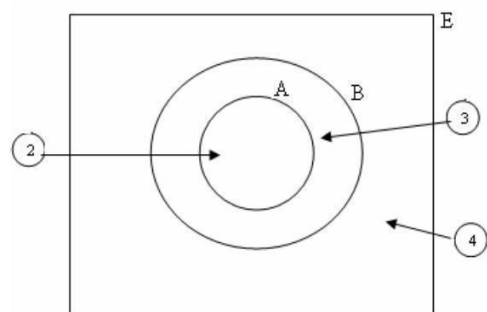
$E: (1), (2), (3), (4)$

$A: (1), (2)$

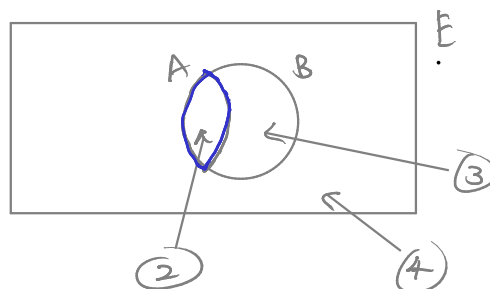
$B: (2), (3)$

Special Venn diagrams for two sets.

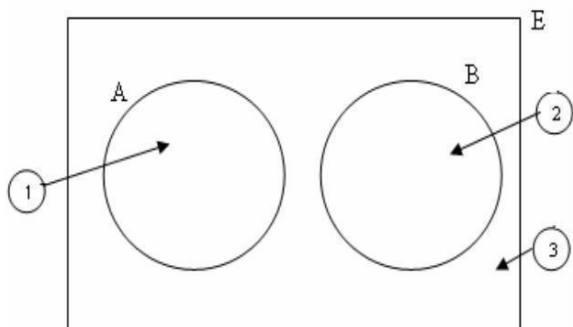
Now, let us consider two sets A, B such that $A \subseteq B$. Then, the Venn diagram for this *special situation* is:



$A \subseteq B$ iff area (1) is empty



Let us consider two sets A, B such that $A \cap B = \emptyset$. Then, the Venn diagram for this *special situation* is:



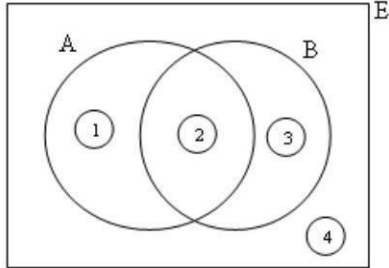
Proof by Venn diagram

Suppose A, B are subsets of a universal set.

Then: (i). $(A \cup B)^c = A^c \cap B^c$

(ii). $(A \cap B)^c = A^c \cup B^c$.

These two equations are called *De Morgan's laws*.



Venn diagram proof:

$E: \textcircled{1}, \textcircled{2}, \textcircled{3}, \textcircled{4} \quad \text{-----}(\alpha_1)$

$A: \textcircled{1}, \textcircled{2} \quad \text{-----}(\alpha_2)$

$B: \textcircled{2}, \textcircled{3} \quad \text{-----}(\alpha_3)$

$A^c: \textcircled{3}, \textcircled{4} \quad \text{-----}(\alpha_4) \quad (\text{by } (\alpha_1), (\alpha_2))$

$B^c: \textcircled{1}, \textcircled{4} \quad \text{-----}(\alpha_5) \quad (\text{by } (\alpha_1), (\alpha_3))$

$\therefore A^c \cup B^c: \textcircled{1}, \textcircled{3}, \textcircled{4} \quad \text{-----}(\alpha_6)$

$A \cap B: \textcircled{2} \quad \text{-----}(\alpha_7) \quad (\text{by } (\alpha_2), (\alpha_3))$

$\therefore (A \cap B)^c: \textcircled{1}, \textcircled{3}, \textcircled{4} \quad \text{-----}(\alpha_8) \quad (\text{by } (\alpha_1), (\alpha_7))$

$A \cap B: \textcircled{2} \quad \text{-----}(\alpha_9) \quad (\text{by } (\alpha_2), (\alpha_3))$

$\therefore (A \cap B)^c: \textcircled{1}, \textcircled{3}, \textcircled{4} \quad \text{-----}(\alpha_{10}) \quad (\text{by } (\alpha_1), (\alpha_9))$

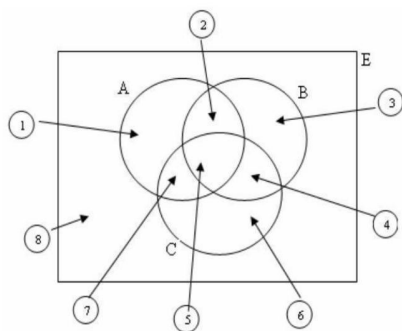
$A^c \cap B^c: \textcircled{4} \quad \text{-----}(\alpha_{11}) \quad (\text{by } (\alpha_4), (\alpha_5))$

By $(\alpha_8), (\alpha_{11}) \quad (A \cup B)^c = A^c \cap B^c$

By $(\alpha_{10}), (\alpha_6) \quad (A \cap B)^c = A^c \cup B^c \quad \square$

3.4(b) Venn diagrams for three sets.

Let us consider three arbitrary sets A, B, C . Then general Venn diagram for this is:



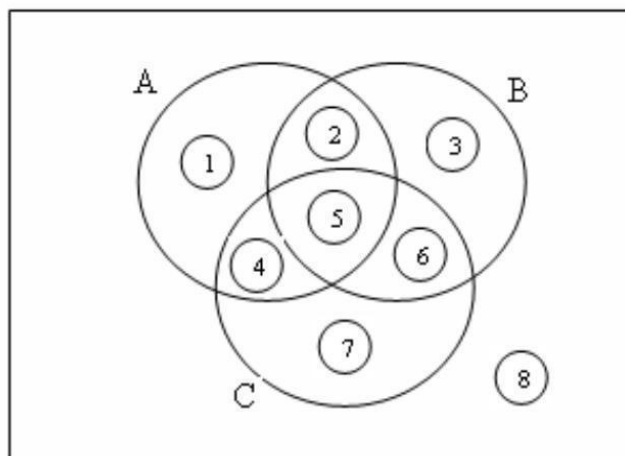
We see that there are *eight* areas. We have labeled them $\textcircled{1}, \textcircled{2}, \dots, \textcircled{8}$.

Notice that, an object x in area $\textcircled{1}$, if and only if $x \in A$ and $x \notin B$ and $x \notin C$. So, area $\textcircled{1}$ corresponds to $A \cap B^c \cap C^c$.

A, B, C are sets. Then:

- (i). $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- (ii). $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Venn diagram proof:



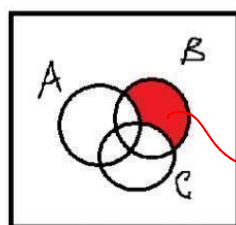
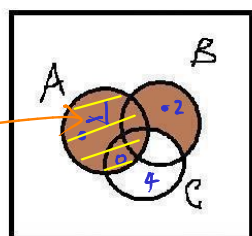
$E: \textcircled{1}, \textcircled{2}, \textcircled{3}, \textcircled{4}, \textcircled{5}, \textcircled{6}, \textcircled{7}, \textcircled{8} \text{ --- } (\alpha_1)$
 $A: \textcircled{1}, \textcircled{2}, \textcircled{4}, \textcircled{5} \text{ --- } (\alpha_2) *$
 $B: \textcircled{2}, \textcircled{3}, \textcircled{5}, \textcircled{6} \text{ --- } (\alpha_3)$
 $C: \textcircled{4}, \textcircled{5}, \textcircled{6}, \textcircled{7} \text{ --- } (\alpha_4)$
 $B \cup C: \textcircled{2}, \textcircled{3}, \textcircled{4}, \textcircled{5}, \textcircled{6}, \textcircled{7} \text{ --- } (\alpha_5)$
 $A \cap (B \cup C): \textcircled{2}, \textcircled{4}, \textcircled{5} \text{ --- } (\alpha_6) \text{ (by } (\alpha_2), (\alpha_5))$
 $A \cap B: \textcircled{2}, \textcircled{5} \text{ --- } (\alpha_7) \text{ (by } (\alpha_2), (\alpha_3))$
 $A \cap C: \textcircled{4}, \textcircled{5} \text{ --- } (\alpha_8) \text{ (by } (\alpha_2), (\alpha_4))$
 $(A \cap B) \cup (A \cap C): \textcircled{2}, \textcircled{4}, \textcircled{5} \text{ --- } (\alpha_9) \text{ (by } (\alpha_7), (\alpha_8))$
 $\text{By } (\alpha_6), (\alpha_9), A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \text{ is proved.}$

Ex.

(a). Prove: $A \cap (B \setminus C) = (A \cap B) \setminus (A \cap C)$.

(b). Give an example to show that $A \cup (B \setminus C) \neq (A \cup B) \setminus (A \cup C)$.

$A \cup (B \setminus C)$



$(A \cup B) \setminus (A \cup C)$

$B \setminus C: \textcircled{2}, \textcircled{3} \text{ --- } (\beta_1)$
 $A \cap (B \setminus C): \textcircled{2} \text{ --- } (\beta_2) \text{ (by } (\alpha_2), (\beta_1))$
 $(A \cap B) \setminus (A \cap C): \textcircled{2} \text{ --- } (\beta_3) \text{ (by } (\alpha_7), (\alpha_8))$
 $\text{by } (\beta_2), (\beta_3), A \cap (B \setminus C) = (A \cap B) \setminus (A \cap C) \text{ is proved.}$

Let $A = \{-1, 0\}$ and $B = \{2\}$, $C = \{0, 4\}$.

$$B \setminus C = \{2\}.$$

$$A \cup (B \setminus C) = \{-1, 0, 2\} \quad \text{--- (i)}$$

$$A \cup B = \{-1, 0, 2\}$$

$$A \cup C = \{-1, 0, 4\}$$

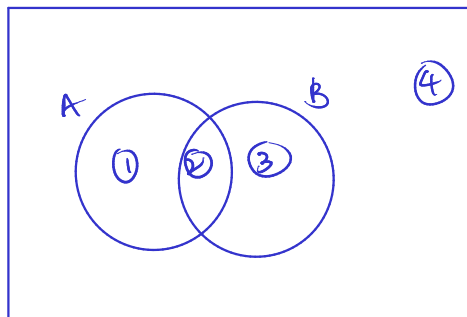
$$\therefore (A \cup B) \setminus (A \cup C) = \{2\} \quad \text{--- (ii)}$$

$$\text{By (i)} \quad A \cup (B \setminus C) \neq (A \cup B) \setminus (A \cup C)$$

i.e. For any sets A, B and C , $A \cup (B \setminus C) = (A \cup B) \setminus (A \cup C)$ is false

Ex.

- (i) Prove that $A \cap B = A$ if and only if $A \subseteq B$. (Using Venn diagrams)
- (ii) Prove or disprove $A \setminus B^c = B \setminus A^c$.



E: $\{1, 2, 3, 4\}$ --- (α_1)

A: $\{1, 2\}$ --- (α_2)

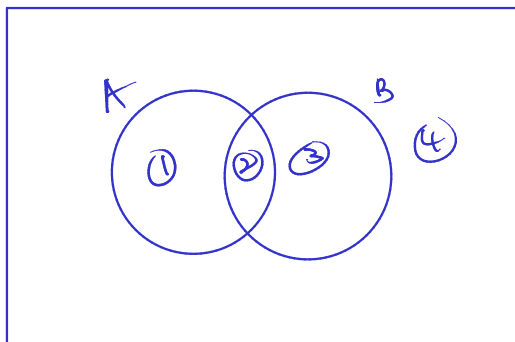
B: $\{2, 3\}$ --- (α_3)

$A \cap B$: $\{2\}$ --- (α_4) (by $(\alpha_2), (\alpha_3)$)

$A \cap B = A$ iff area 1 is empty --- (α_5)
(by $(\alpha_2), (\alpha_4)$)

$A \subseteq B$ iff area 1 is empty --- (α_6)
(by $(\alpha_2), (\alpha_3)$)

by $(\alpha_5), (\alpha_6)$, $A \cap B = A$ iff $A \subseteq B$ $\square$



E: $\{1, 2, 3, 4\}$ --- (α_1)

A: $\{1, 2\}$ --- (α_2)

B: $\{2, 3\}$ --- (α_3)

B^c : $\{1, 4\}$ --- (α_4) (by $(\alpha_1), (\alpha_3)$)

$A \setminus B^c$: $\{2\}$ --- (α_5) (by $(\alpha_2), (\alpha_4)$)

A^c : $\{3, 4\}$ --- (α_6) (by $(\alpha_1), (\alpha_2)$)

$B \setminus A^c$: $\{2\}$ --- (α_7) (by $(\alpha_3), (\alpha_6)$)

$\therefore A \setminus B^c = B \setminus A^c$ $\square$

Prove that $A \setminus B^c = B \setminus A^c$ using their definition.

Proof:

Let x be an arbitrary element. --- (i)

$$x \in A \setminus B^c \Leftrightarrow x \in A \text{ and } x \notin B^c$$

$$\Leftrightarrow x \in A \text{ and not } (x \in B^c)$$

$$\Leftrightarrow x \in A \wedge \neg (x \in B^c)$$

$$\Leftrightarrow x \in A \wedge (\neg (x \in B^c) \vee x \in B)$$

$$\Leftrightarrow x \in A \wedge (F \vee x \in B)$$

$$\Leftrightarrow x \in A \wedge x \in B$$

$$\Leftrightarrow x \in A \cap B$$

$$\therefore A \setminus B^c = A \cap B \text{ --- (i)}$$

$$(C = D \Leftrightarrow \forall x, x \in C \Leftrightarrow x \in D)$$

$$C = D \Leftrightarrow C \subseteq D \wedge D \subseteq C$$

$$x \in B \setminus A^c \Leftrightarrow x \in B \wedge x \notin A^c$$

$$\Leftrightarrow x \in B \wedge \neg (x \in A^c)$$

$$\Leftrightarrow x \in B \wedge \neg (x \in A^c \vee x \notin A)$$

$$\Leftrightarrow x \in B \wedge (x \notin A^c \vee x \in A)$$

$$\Leftrightarrow x \in B \wedge x \in A$$

$$\Leftrightarrow x \in A \cap B$$

$$\Leftrightarrow x \in A \cap B$$

Similarly $B \setminus A^c = B \cap A$
 $\stackrel{?}{=} A \cap B \text{ --- (ii)}$

$$\begin{aligned}
 A - B^c &= A \cap (B^c)^c \\
 &= A \cap B \\
 &= B \cap A \\
 &= B \cap (A^c)^c \\
 &= B - A^c
 \end{aligned}$$

(R1)
 (R2)
 (R3)
 (R2)

R1: $A - B = A \cap B^c$
 R2: $(A^c)^c = A$
 R3: $A \cap B = B \cap A$

End of D/6